ANSIBLE

Automated: Deployment, Installation [ONE AT A TIME]

IT IS FREE

IT IS OPENSOURCE

IT IS PLATFROM INDEPENDENT

IT IS VERY IMP TOOL IN DEVOPS

To perform end-to-end automation we can use Ansible.

Creating servers

configure servers

deployment application on servers

Not with single, it can deal with multiple server.

ANSIBLE ALTERNATIVES: CHEF, PUPPET, SALTSTACK ----

ANSIBLE:

it's a Configuration Management Tool.

Configuration: Hardware and Software

Management: Packages update, installing, remove ----

Ansible is used to manage and work with multiple servers together.

it's a free and Opensource.

it is used to automate the entire deployment process on multiple servers.

We install Python on Ansible.

we use a key-value format for the playbooks.

PLAYBOOK:

create servers

install packages & software

deploy apps

Jenkins = pipeline = groovy

ansible = playbooks = yaml

YAML: YET ANOTHER MARKUP LANGUAGE

HISTORY:

in 2012 dev called Maichel Dehaan who developed ansible.

After few years RedHat taken the ansible.

it is platform-independent & will work on all Linux flavours.

ARCHITECTURE:

PLAYBOOK: it's a file which consist of code

INVENTORY: it's a file which consist Ip of nodes

SSH: used to connect with nodes

Ansible is Agent less.

Means no need to install any software on worker nodes.

MASTER-SLAVE CONCEPT :

STEP-1: LAUNCH 3 INSTANCE (1-MASTER, 2-SLAVE)

STEP-2: INSTALL ANSIBLE, PYTHON AND PIP ON MASTER SERVER

yum install ansible -y

yum install python-pip -y

ADD INVENTORIES (vi /etc/ansible/hosts) ON MASTER SERVER

```
# Ex 2: A collection of hosts belonging to the 'webservers' group
[dev]
172.31.34.110
172.31.35.94

[test]
172.31.38.217
172.31.40.252
## [webservers]
```

STEP-4 : GENERATE A KEY IN ROOT USER ON MASTER SERVER (ssh-keygen)

OPEN SLAVE SERVER AND FOLLOW THE BELOW STEPS:

STEP-5: SET A PASSWORD TO USER IN ROOT SERVER (passwd root)

STEP-6: NOW WE HAVE TO SAY YES TO PASSWORD AUTHNETICATION

vi /etc/ssh/sshd_config ----> 65 line (63gg)

```
PasswordAuthentication yes
```

change the password authentication from **no to yes**

line number 40 : **remove #** and type **yes**

Ex: PermitRootLogin yes

STEP-7: RESTART SSHD (systemctl restart sshd)

MASTER SERVER AGAIN AND COPY THE PUBLIC KEY:

STEP-8: COPY THE PUBLIC KEY TO ALL SLAVE SERVERS (ssh-copy-id root@slave_ip)

STEP-9: TO CHECK WITH SLAVE SERVER CONNECTION to check the connection : ansible all -m ping

FIRST PLAYBOOK:

```
---
- hosts: dev
  connection: ssh
  tasks:
  - name: installing git on dev server
    yum: name=git state=present
```

To execute the playbook : ansible-playbook filename